



Typography's Effect on Language Learners' Reading Processes: An Eye Tracking Study

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Bioprofile

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Abstract

Typography, in shaping how language appears, functions as the element of design that most influences the way people access a text to extract meaning. One of the many roles EFL educators assume is that of a materials developer, creating texts which are then distributed and consumed by language learners. All texts inherently contain typographical cues and hierarchy that may influence reactions to and processing of texts. This paper reports the findings of a study designed to investigate the effects of typographical cueing and hierarchy on the reading patterns, recall, and comprehension of participants whose first language is not English. Participants were split into two groups: The control group was exposed to an 'undersigned' document, while the experimental group read a 'designed' version of the same document containing typographical hierarchy and cueing. Participants' reading process was measured by an eye tracking camera. Participants completed comprehension and recall questions post reading. The results show significant differences on recall and comprehension tasks, with the experimental group performing better than the control group. Eye tracking data also showed differences between the two groups regarding fixations and reading patterns

used within the texts. This study shows how typography influences both access to and comprehension of texts read by language learners. The results provide support for the importance of well-designed materials and should encourage EFL educators, whether design savvy or not, to focus more on the typography of their materials.

Introduction

Language surrounds us and permeates our environment in many forms, both auditory and visual. Language, in its various forms and modes, is partially duplicitous in its nature, “some things can be ‘said’ only visually, others only verbally” (Kress & van Leeuwen, 2006, p. 2). Though language is split into these modes, the process of extracting meaning from language, both auditory and visual, forms the basis for communication. In the case of visual communication, the appearance of the medium itself plays a role in meaning making. Typography, “what communication looks like” (Felici, 2012, p. ix), is to visual communication what register or phonology are to verbal communication. Candlin (2001) promotes typography as a “central part” and “key point of access to the rhetoric of a range of subjects” (p.1). Typography also has an effect on our engagement with a text (p.1). Lupton (2010) elucidates this idea further by providing insight into the process of typographical design in this manner:

Designers provide ways into-and out of-the flood of words by breaking up text into pieces and offering shortcuts and alternate routes through masses of information...one of design's most humane functions is, in actuality, to help readers avoid reading. (p. 87)

There are many methods of designing a text that can effectually enable typography to guide the reader through a text (see Beck, 1984; Foster 1979). Walker (2001) states “the role of a typographer [is to] articulate the meaning of a text, making it easy for readers to understand” (p. 3). Second language acquisition and education is an ideal field where “shortcuts and alternate routes” that facilitate ease of understanding to play a key role. Logic would suggest that typography could be used to expedite the process of extracting meaning from a text. However, empirical data is needed to prove that typography directly influences both the reading process and post-reading recall. There have been only few studies investigating the effect of typography on the reading process, and none that specifically focus on second language learners. The current study combines traditional recall measures with state of the art eye tracking, in a novel approach to measure how typography can alter language learners’ reading processes and ability to recall details. This study’s experimental design involved two

groups exposed to two different typographical variations of a text as a basis for measuring the relationship of typography, recall, and reading patterns. The two variations of the text were lexically identical, but with typographical modifications put into place for the experimental group. Participants' gaze was measured by an eye-tracking camera and participants were tested on their recall of the passages. In this manner, the study was designed to fill the gap in our knowledge of how typography, as an element of design, affects the way language learners interact with print or digital materials. Analysis of the data suggests that language learners' comprehension and reading patterns are indeed altered by the application of typographical design principles, contributing to our understanding of learners' reading processes and interactions with texts.

Literature Review

Texts may serve a variety of roles in the EFL classroom. Sometimes, instructional texts may be used to give learners instructions or guidance, while at other times, texts will be examined intensively and in great detail. First, this review will explore typography's role in the literature of supporting texts in fulfilling their purpose and outcomes. Following this, research concerning typographical cueing and hierarchy will be presented as a way to show how classroom texts can be enhanced by using typographical principles. Finally, the literature on eye tracking, reading, and typography will be covered to show how technology can provide insightful data into the complex reading process.

Typography, Readability, and Legibility

Typography is intimately connected with the process of reading texts that are designed to fulfill a wide range of purposes. Therefore, for each textual type and audience, typography is modified to best serve its capacity of supporting the meaning and purpose of the text. There are two key concepts that inform the typographical style: readability and legibility. According to Williams (2006, p. 33), "Readability refers to whether an extended amount of text—such as an article, book, or annual report—is easy to read." A novel or short story would be examples of texts that need to be readable. "Legibility refers to whether a short burst of text...a headline, catalog listing, or stop sign—is instantly recognizable" (Williams, 2006, p. 33). In our daily encounters with texts both in print and online, we may spend a substantial portion of our reading time scanning the text quickly. Legible text allows us to quickly pull relevant information from a text. As Lupton (2010) commented, "One of design's most humane functions is, in actuality, to help readers avoid reading" (p. 87). Typography can

enhance either the legibility or readability of a given text to better fulfill its role as a communicative medium.

Typography and Reading Fluency

Whether a text is designed for legibility or readability, the time spent interacting with a text is an important measure of a text's efficiency in communicating a message. Many researchers have thereby focused on the relationship of typography and reading fluency. A great majority of these studies use four key typographical variables, typeface, point size, spacing, and line length (Papadopoulou, Manoli, & Zifkou, 2014, p. 24) and measure their influence on reading rate. Beymer, et al. (2008), for instance, found a "slight reading speed advantage for larger fonts" with longer fixation periods on smaller fonts (p. 16). Thiessen and Dyson (2010) conducted a similar study concentrated on type size and spacing's impact on reading performance, albeit with children as participants (N=6). "Findings indicated that the performance of the children varied based on the size and spacing between the two groups" (p. 371). Specifically, the children performed better with the smaller sized texts. Dyson and Haselgrove also indicate that line length can affect reading rate, which in turn affects performance on comprehension measures (p. 589). These studies suggest appropriate manipulation of font size and line length can be used to slow down or speed up the reading process, allowing materials developers to draw attention and reader resources to important textual details.

The aforementioned study by Beymer, et al. (2008) is one of the only studies in the literature to study if having English as a first language makes a difference on the reading process. Their results indicate that the "non-English first group" read at a predictably slower rate, and re-read the text significantly more times. However, other than the time considerations, the non-English first group performed equally well on comprehension measures (p. 18).

Typographical Cueing and Education

Typeface, point size, spacing and line length are all holistic typographical features of a text. Generally, researchers have maintained the internal consistency of a text by keeping these variables the same throughout. Aside from these more holistic variables, there are other typographical features used to enhance either readability or legibility. These are known as typographical cues, which are used "to signal the important ideas in a text" (Waller, 1991, p. 245). Typographical cueing focuses on the more granular details of type, such as weight, style, or color, rather than the more holistic hierarchy of the entire document. Research by

Papadopoulou, et al. (2014) established that even preschool age children notice typographical cues (p. 34).

Jourdenais, et al. (1995) demonstrated that typographical cues can move beyond noticing to integration and application. They enhanced a text by highlighting (a form of typographical cue) preterit and imperfect verb forms. After reading, participants were given a language production task. The treatment group's productive language was found to contain more of the target typographically-cued structures (p. 183). Typographical cueing in this case was found to have a direct impact on post-study language production. These results are in line with Coles and Fosters' (1975) observation that "people are more likely to remember cued ideas" (p. 106), but expand this by showing that cueing can extend its influence to language production.

Waller (1991) notes that "[a]nything about a text which is discernable to readers may affect their perception of the status of a document and consequently their expectations, critical stance, reading strategies, goals, and outcomes" (p. 344). Authors of a given text may desire a specific outcome following interaction with their text. Educators, for example, may desire to improve the recall and uptake of certain key information. Lorch, et al. (1995) studied this by creating 3 different versions of a text with cueing ranging from none (the control group), light, and heavy. Results indicate that "[c]ued recall was better in the light condition than in the control or heavy conditions, which did not differ" (p. 51). Excessive use of cueing can normalize the cues, minimalizing their impact on retention. Lorch, et al. found that capitalizing target words, a rather simple cue and feature of the 'light' passage, slowed the reading process and increased participant's ability to recall the targets (p. 51).

Typographical Hierarchy

Typographical cues provide important visual distinction in a text that makes it easier to navigate by establishing a hierarchical structure within a text. Depending on the language, text flows in numerous ways, punctuated by entrance and exit points into chunks of text. For example, English text flows right to left, top to bottom, generally with a number of breaks that signify idea units or groupings. Typical texts have an established hierarchy to guide the reader through a text. Lupton (2010) explains, "A typographic hierarchy expresses the organization of content, emphasizing some elements and subordinating others" (p. 132). This process of emphasis and subordination takes place using typographical cues such as headings, spacing, line length, and indentation.

In addition, there are some guiding principles that help establish a clear hierarchy in a text. The design principles of contrast, alignment, proximity, and repetition (see Williams, 2006, p. 13) can be applied to a text to establish a clear hierarchy of display text, titles, headings, subheadings, and body text, furthering the ease of reading a document. Bergstrom and Schall (2014) also endorse using headings and subheadings to construct a clear hierarchy (p. 167). Canning (2004) notes that considering design principles helps to “display the overall structure of the text by visibly segmenting the text into distinct sections” (p. 2). A text with a clear flow created by typographical hierarchy is easier for any reader to access.

Eye-Tracking Studies

Measuring the effect of typography on the reading process is an arduous process. Reading, including reading in a foreign language, is an intricate harmony of complex thought patterns and minute optical muscular movements. Our understanding of this multifaceted process has been enhanced in recent years by the advent of eye-tracking technology. This technology allows us to track the complex patterns of eye movement. Many studies have been completed in the fields of psychology and linguistics, revealing the way our eyes move through text. For the purpose of this study, we’ll examine eye tracking research’s contribution to knowledge concerning how the nature of the text affects the reading process.

Godfroid, et al. (2013) used eye-tracking to investigate whether increased attention leads to enhanced learning. Their study used eye tracking to measure participants’ attention to different words. They found that the more participants fixated on words, the more they were able to recall vocabulary on a posttest (p. 484). This study, though not focused on the design of the text itself, supports the noticing hypothesis and the idea that as content becomes more salient, it becomes easier to recall.

Interpretation of a text is inherently subjective, including judgments based on the general appearance of the text. Rello and Marcos (2012) investigated readers’ preferences for specific textual features and compared these with eye tracking data to ascertain if features actually were effective in drawing reader’s attention. In summary, the researchers stated that, “Text customization has an impact on readability. At the same time, some textual layouts are preferred to others regarding reading comfort.” (Rello & Marcos, 2012, p. 64). One feature that had an impact on readability was font size, the smallest size, 14pt, was the least preferred size, but increased the duration of fixation on the text.

Furthermore, the words of a text themselves have been found to affect reading in one’s native language. Traxler and Pickering (1996) used eye tracking to measure processing of

plausible and implausible grammar structures and found that in improbable cases, the fixation time was considerably longer (p. 461). Bergstrom and Schall (2014), provide a concise summary of the features of written language that lead readers to fixate on or notice particular words. The type of word, content or function, and even word length influence reading patterns (see Carpenter & Just, 1983; Rayner & McConkie, 1976). While the words themselves were not typographically cued, these studies show how the application of the eye tracking can provide information on how even minute details of text influence reading.

However, all of the aforementioned studies have focused on reading in one's first language. Beymer, et al. (2008) is one of few studies to research the effect of typeface, point size, and spacing using eye-tracking and employing participants with a first language other than English. Their study showed that participants whose first language was not English approached the text in a different manner, with more rereading, but achieved similar results on the comprehension measures (p. 18).

While there is substantial evidence that reading skills transfer from learned languages to additional languages (see Roberts, 1994), there is a need for empirical data to support the application of the findings in the literature to language learners. The current study seeks to fill this gap by focusing on English language learners and their exposure to educational texts.

Rationale and Research Questions

Research indicates that typography can greatly influence reading and post-reading performance. Typographical cueing, hierarchy, line-spacing, typeface, point size, word choice, all influence the reading process. There is a shortage of research regarding the effect typography has on language learners. In addition, empirical data is needed to connect typography, the reading process and post reading recall. This study uses two versions of text, one 'undesigner' and the other 'designed' with typographical cueing and hierarchy to increase the salience of valuable information. Rather than focus on typeface, point size, or line spacing as has been done in many past studies, the researcher focused on the use of typographical cueing and hierarchy. Using two versions of a text, an eye-tracking camera, and a post-reading quiz, the researcher focused on these research questions:

- 1 Does the inclusion of typographical cueing in a text lead to increased comprehension and recall of a text?
- 2 Which hierarchical or typeface related cues are most effective at drawing the attention of participants?

3 How do typographical cues influence the way participants access a text?

Methodology

Passage

The researcher created a passage on phonology for the test passage, due to students' unfamiliarity with the subject matter and small likelihood of completing comprehension tasks based on prior knowledge. The researcher was careful to include enough thought groups with key points that could be used for recall tasks. The original version of the text was approximately 200 words in length. Following pilot testing, the length of the passage was reduced to 140 words to focus more on the essential information. The typeface, Helvetica at 12-point size, was selected for two reasons: First, the ubiquity and familiarity of the typeface, especially for participants accustomed to smartphone use, and second, researchers have noted Helvetica's neutrality and lack of distraction (see Itkonen, 2006; Waller, 2011, p. 8). Leading (line spacing) was set at exactly 20 points. Line length was set at 157mm. All of the features above were used to create both the undesigned and designed passages. Both passages were not only identical lexically, but also the typeface, base point size, line length, and line spacing were controlled. See Appendix A for the baseline, undesigned passage (U).

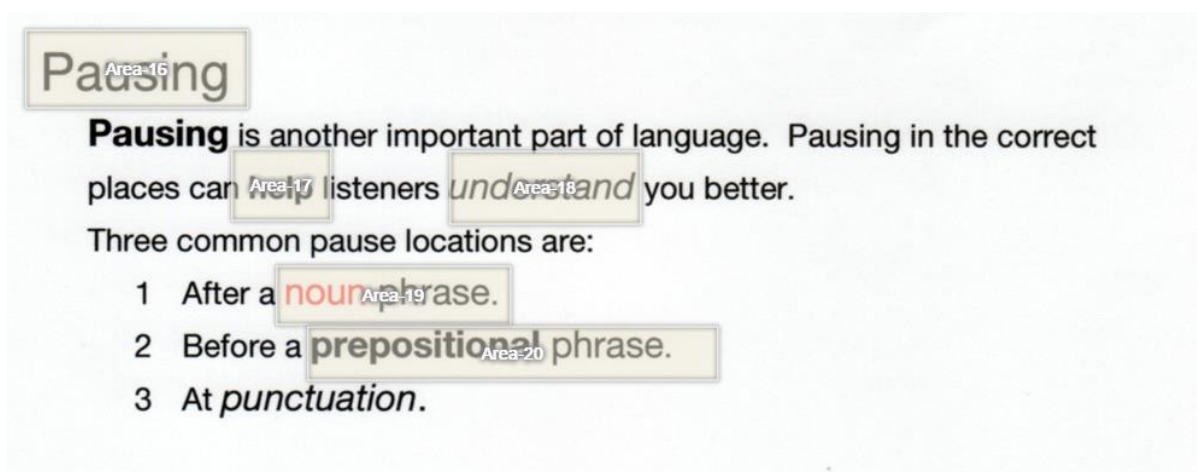


Figure 1. Sample of Areas of Interest Groupings for the Designed Passage

The designed passage was then altered to include typographical and hierarchical cueing. First, hierarchy was created by increasing the point size of titles and headings. Cues were added to increase the salience of important words. These cues included color, bold, italics, baseline shift, and size adjustments (but not altering the base point size). See Appendix B for the designed passage (D). There were a total of 21 changes made to the

designed document. For eye tracking purposes, each of these changes was selected in the eye-tracking software as an Area of Interest (AOI). The corresponding areas in the undesigned document were also marked as AOIs so that comparative analysis could be undertaken.

Participants

Participants were selected from a pool of Japanese 2nd and 3rd year university students ($N=65$) majoring in English. All participants had achieved a TOEIC score of 480 or higher at the time of completing the study. None of the students had lived for a considerable time (more than 6 months) outside of Japan. Their ages ranged from 19-21 years old and there was a mix of both male and female students.

Procedure

Participants were randomly assigned to the control or experimental group. Participants were compensated at the student employee rate for their time. Participants completed the tasks individually one-by-one. First, participants were led into a quiet, controlled environment to a computer with an eye tracker attached. Participants read and signed an informed consent form in their native language of Japanese. They were then presented with a simple set of instructions, also in Japanese.

Participants were oriented in front of a monitor with a resolution of 1920x1080, set at 80% brightness. An eye tracking camera was set below the display. This eye tracker, The Eye Tribe Tracker from The Eye Tribe, features a sample rate of 30-60 Hz and accuracy of 0.5-1 degree. Participants were oriented in front of the tracker at a fixed distance. Participants then completed a calibration process to ensure the tracker accurately measured eye movement. Following calibration, participants were presented with either the designed or undesigned text. The text was displayed for 70 seconds, due to feedback from pilot testing. Post reading, participants completed a set of comprehension and recall questions on the computer. These questions included both multiple choice and open-ended response items.

Analysis

Eye-tracking data was compiled using the Eye Proof Suite of tools for research and analyzed using Sofa Stats statistical analysis software. The eye-tracking camera was unable to track several participants' eye movements, so these students' data were removed from the study. Data analysis was completed for 23 participants exposed to the designed passage and 22 participants exposed to the undesigned passage ($N=45$). Areas of Interest (AOI) were

normalized according to size and analyzed for fixation differences between the two versions of the passage.

Comprehension and recall items were assigned numerical values. Open-ended items were scored using a rubric. This data was also analyzed using Sofa Stats statistical analysis software.

Results

Eye-tracking Data

There was no significant difference in the average fixations of members of both groups. This indicates that participants were attentive to, and actively tried to read their version of the document within the time frame.

Percent fixated is a measure of how many participants focused on a particular area of interest (AOI). The average value for percent of AOIs fixated was higher for the designed group (D) at 67.38% than the undesigned group (U) at 64.66% [$t=0.193$, $df=43$, $p=0.8482$], though not significant. However, there were specific AOIs that showed greater variance between groups. Specifically, AOI 6, which was fixated by 100% of D group participants and only 62.07% of U group participants [$t = 3.273$, $df = 43$, $p = 0.0021^*$]. Area 6 contained multiple typographical cues, including size, baseline shift, and boldness.

Other AOIs from the designed passage (D), 1, 8, 10, 11, 12, 13, 14, and 21 all were fixated in excess of 10% more than their undesigned counterparts, though these were not found to be significant. Interestingly enough, all of the above areas, with the exception of area 11, contained multiple typographical cues, e.g. color and size.

On the other hand, AOIs 4, 17, 19, 20 were fixated in the undesigned passage at a rate greater than 10% more than they were in the designed passage. AOI 4 occurred at the beginning of a line, and areas 19 and 20 were part of a list, which are both examples of simple typographical hierarchy present in the undesigned text.

Averaged reading patterns showed both groups began the task by fixating on the center of the document, following which they quickly moved to the top of the page. The designed group focused more on the headings and large words, while the undesigned group seemed to focus more on the beginning of each line text, relying on their knowledge of rhetoric to guide them through the text.

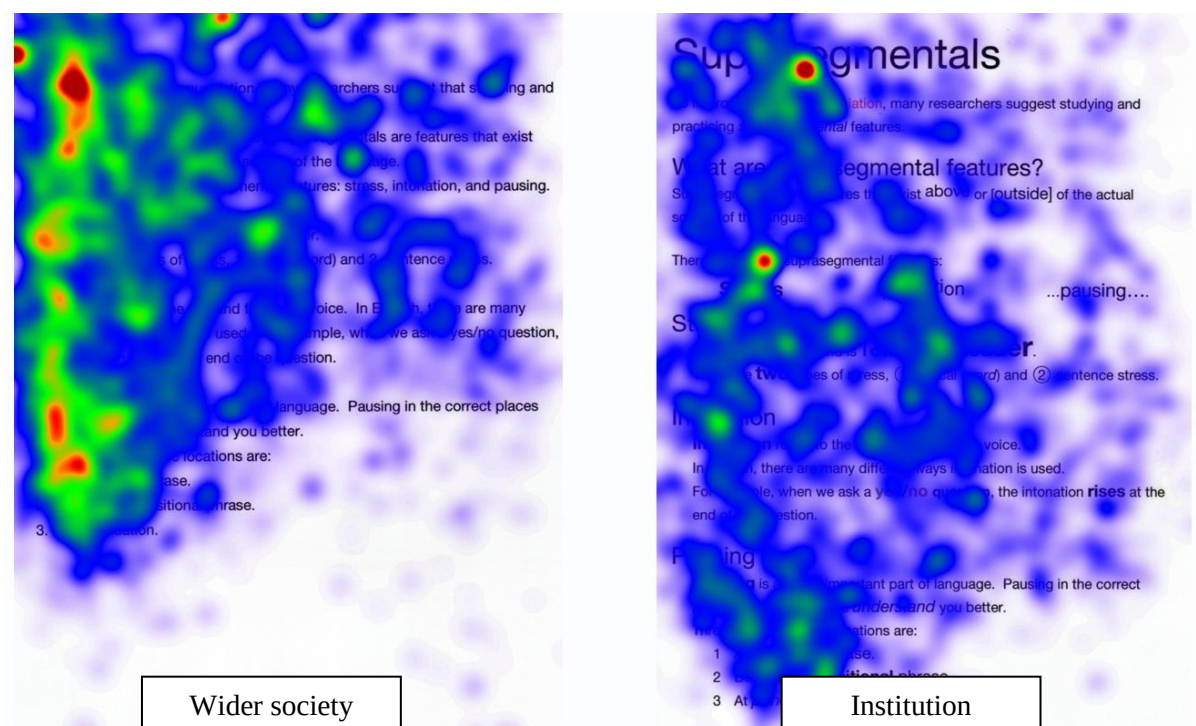


Figure 2. Comparison between heatmaps for the designed and undesigned texts.

Comprehension and Recall

Due to the normal nature of the data, with participants randomly chosen out of homogenous pool, a t-test was determined as the best statistical measure to determine differences between groups. Participants' total scores on the posttest measures were analyzed using a t-test (see Table 1). The analysis showed significant difference between the two groups.

Table 1.

T- test of Total Comprehension Recall Scores

Group	N	Mean	SD	Min	Max	T
Designed	23	8.78	2.80	4.0	13.0	3.532
Undesigned	22	5.68	3.09	1	12.0	

$p < 0.001^*$

Degrees of Freedom (df): 43

An ANOVA was also performed on the total scores, to provide further information about the nature of variance between the two groups.

Table 2

ANOVA for Designed and Undesigned Groups

Source	SS	Df	MS	F	P
Between	108.114	1	108.114	12.474	< 0.001*
Within	372.686	43	8.667		

Similarly, the ANOVA indicated that there was indeed significant variance between the two groups.

Next, a *t*-test (see Table 3) was performed on the total score for the multiple-choice questions.

Table 3

T- test of Multiple-Choice Questions

Group	N	Mean	SD	Min	Max	T
Designed	23	2.83	0.83	1	4.0	3.57
Undesigned	22	1.82	1.05	0	4.0	3.57

$p < 0.001^*$

Degrees of Freedom (*df*): 43

A significant difference was observed between the two groups. The designed group's scores higher than the undesigned group. Additionally, a *t*-test of participants' scores on the open-ended recall items (Table 4) showed similar results, albeit with a slightly higher *p* value.

Table 4

T- test of Open-Ended Questions

Group	N	Mean	SD	Min	Max	T
Designed	23	5.96	2.67	2	10.0	2.62
Undesigned	22	3.86	2.70	0	10.0	2.62

$p = 0.01223$

Degrees of Freedom (*df*): 43

Additional analysis in the form of *t*-tests, was run for each individual item on the comprehension text, in an effort to isolate areas of interest linked to typographical cueing.

Question 3, linked to Area of Interest 4, was found significant at a p value of 0.05 [$t = 1.974$, $df = 43$]. Area of Interest 4 contained a size adjustment and baseline shift. Question 5 reached a higher level of significance [$t = 3.287$, $df = 43$, $p = 0.002$] and was linked with AOI 5 and 6, cued by boldness, baseline shift, and size. Question 10 [$t = 2.996$, $df = 43$, $p = 0.004$] was linked with AOI 13, which was cued using size and baseline shift. All questions with significant differences were linked to AOI enhanced by size and baseline shift, although it cannot be concluded whether this relationship is causal or merely correlational.

Discussion

Research Questions

Does the inclusion of typographical cueing in a text lead to increased comprehension and recall of a text?

Typographical cueing does seem to influence comprehension and recall of information from a text. However, as the comprehension test was taken directly following interaction with the text, this supports the findings of Coles and Foster (1975) that typographical cueing will lead people to remember cued ideas and improve immediate recall. Language learner's ability to immediately recall information is an area of concern for classroom practice. If the saliency of information is enhanced through typography, the avenue of spaced review and repetition are available to help cement ideas and concepts in learners' longer term memories. The inclusion of typographical cuing opens the doors to deeper learning of material.

The results of this study also indicate that the nature of typography itself contributes to uptake of materials. The presence of multiple typographical cues in significant recall items suggests that noticeably visibly altered portions of text draw the attention of the reader and enhance recall ability. Nevertheless, the inclusion of too much typographical cues may also desensitize learners to cuing, detrimentally impacting the ability of students to recall key ideas from the text. An optimal amount of typographical cues has not been determined by prior research, but it may be possible to posit an answer based on the data collected for this study. The data indicate significant difference in recall when one to two typographical cues are added per paragraph. Educators would do well to think of what information they most want their students to remember and apply typographical cues to enhance the saliency of that text.

Which hierarchical or typeface related cues are most effective at drawing the attention of participants?

Creating a hierarchy of importance from the title to the headings to the body text using contrasts in typeface size seem to have the largest influence on drawing the attention of participants. The design principle of contrast is clearly at work here. To put it simply in the words of Robin Williams, “To make a contrast of size work effectively, don’t be a wimp” (2006, p. 188). Contrasts in type size are one of the simplest ways to show difference between body text and key phrases or words. In this case, statistical difference was found when using text altered by two to eight points in size over the base text size. Headings were sized at 20 points compared to the base font size of 12 points. The difference between texts created a clear hierarchy and flow of information. Typographical cueing using point size can enhance the hierarchy and the salience of important information in the text.

Furthermore, the data indicate that using baseline shift to interrupt the homogeneity of a line seems to draw attention to lexical items. In this case, this technique is not widely employed in normal texts, therefore instantly recognizable by readers for its peculiarity. For readability purposes, breaking the baseline is not generally recommended as it can potentially slow down the reader. On the other hand, if an educator really wishes to draw attention to an item in a text, by breaking the flow of the text, they can immediately force the reader to spend time processing the shifted text, bolstering it in their memory.

In addition, the data indicate that when more than one typographical cue is applied to text, such as a change in font style and size, language learners are more likely to notice and fixate on the altered text. This finding is in accordance with William’s (2015) principle of contrast, which she succinctly summarizes thus, “If two items are not the same then make them really different...Don’t be a wimp” (p. 65). As language educators and materials developers design texts for language learners, they should consider the ways they can aid uptake and recall of the materials and apply typographical cues to reach the intended objectives of the text.

How do typographical cues influence the way participants access a text?

Other than the reasons mentioned above, analyses of the eye tracking data indicate that texts featuring typographical cueing and hierarchy allow the reader to approach a text in different ways. Returning to words of Lupton (2010) concerning the nature of typography, “Designers provide ways into-and out of-the flood of words by breaking up text into pieces and offering shortcuts and alternate routes through masses of information” (p. 87). By reinforcing the hierarchy of the text and inserting typographical cueing, the researcher created different

access points into the text of the designed version. The heat maps for both passages support this by showing a much heavier focus on conventional top-down left to right reading patterns for participants who read the undesigned passage. The designed passage, with its more developed hierarchical structure featured many access points for participants. Each heading allowed the participants to jump in and access a topically-relevant block of text. Due to this fact, the participants were able to read in a less linear structured fashion. Returning to prior information was easier in a text with more positional cues. The potential for rereading and review is strengthened when texts offer visual differentiation.

On the other hand, before educators decide to employ copious amounts of cueing to all the texts they use in their classes, it is vital to consider the nature of the text. Due to the fact that typographical cues will alter reading patterns, cueing in the manner described in this study may not always be an appropriate choice. First, educators need to consider whether the main purpose of the text is for readability or legibility, a distinction which leads into potentially disparate design philosophies. Once this has been determined, the post-reading outcomes should be determined. Is immediate recall important? Will the meaning of the text be clarified through other mediums? Only then can the degree to which typographical cues are inserted into a text be decided.

Limitations/Suggestions for Future Research

Though this study provides evidence to support the idea that typographical cueing leads to increased memory and recall, it cannot be interpreted to mean that retention of ideas is enhanced. This study primarily focused on short-term recall.

Second, it is difficult to make conclusions about the impact of different typographical features – since the study did not isolate for any particular typographical cues. Further work would need to be conducted to isolate specific typographical features.

Additionally, participants were all of a Japanese language background, therefore the results of this study could have limited applicability to a wider population. Furthermore, a larger sample size would perhaps yield more informative results.

Conclusion

The data provided by this study helps demonstrate this important connection and the potential benefits of design for second language acquisition and education. Simple design provides ways to enhance and make texts more suitable for achieving their intended purpose. Learners are exposed to a myriad of texts containing a wide variety of text types, aesthetics, and genres. Educators can enhance their materials through typography and design for not only

aesthetic purposes, but to improve the pedagogical impact of the texts. Typography and design play a demonstrable part in positively or negatively influencing the way learners interact with a text. In the field of education, where every innovation, ideology, and practice is scrutinized and critiqued to ascertain the best way to teach and learn, design and typography cannot be ignored as they are the gatekeepers of textual-based interaction. Design ushers language learners to important portions of the text and shepherds them from one idea to the next while pointing out important landmarks on the way. All educators would benefit from considering the way how typography and design can positively enhance the amount of material noticed and processed by language learners. Typography is one way to effectually create “an increase in efficiency in how knowledge is communicated” (Lanham, 2006, p. 92).

Language learners are also becoming increasingly design savvy and tend to be more aware of design problems that may influence their learning process. Online culture has created avenues for sharing and discussing educational materials. Classroom materials are often uploaded, critiqued or mocked for their questionable design. Language learners should expect good design that doesn’t impede or hinder their language acquisition, but instead enhances it by providing paths of access, hierarchy, and cues. In ideal circumstances, language learners take ownership of the design of texts and work in conjunction with educators to develop materials that will be best suited for them. This aligns perfectly with Murphey, et al. (2009) assertion that incorporating student voice into ELT practices improves educational quality. The present study adds empirical data to support both institutional and grassroots application of design to materials development and argues for more integration between the fields of design and education. As design and typographical principles make their way into every teacher’s classroom and into international conferences and forums we will see a more effective dissemination of ideas that allows for better teaching and, as a result, more effective learning.

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Appendix A

Undesigned Passage

Suprasegmentals

To improve English pronunciation, many researchers suggest that studying and practicing suprasegmental features.

What are suprasegmental features? Suprasegmentals are features that exist above or [outside] of the actual sounds of the language.

There are three main suprasegmental features: stress, intonation, and pausing.

Stress

Stress means a sound is longer and louder.

There are two types of stress, 1. lexical (word) and 2. sentence stress.

Intonation

Intonation refers to the rise and fall of the voice. In English, there are many different ways intonation is used. For example, when we ask a yes/no question, the intonation rises at the end of the question.

Pausing

Pausing is another important part of language. Pausing in the correct places can help listeners understand you better.

Three common pause locations are:

1. After a noun phrase.
2. Before a prepositional phrase.
3. At punctuation.

Appendix B

Designed Passage

Suprasegmentals

To improve **English pronunciation**, many researchers suggest studying and practicing **suprasegmental** features.

What are suprasegmental features?

Suprasegmentals are features that exist **above** or [outside] of the actual sounds of the language.

There are **3** main suprasegmental features:

Stress

intonation

...pausing....

Stress

Stress means a sound is **longer** and **louder**.

There are **two** types of stress, ① lexical (*word*) and ② sentence stress.

Intonation

Intonation refers to the **rise** and **fall** of the voice.

In English, there are many different ways intonation is used.

For example, when we ask a **yes/no question**, the intonation **rises** at the end of the question.

Pausing

Pausing is another important part of language. Pausing in the correct places can **help** listeners *understand* you better.

Three common pause locations are:

- 1 After a **noun** phrase.
- 2 Before a **prepositional** phrase.
- 3 At *punctuation*.